Mohamed Alaa | Resume

AI/ML Engineer · 6th of October, Egypt
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- Job Status: AI/ML Engineer at Zetaton
- Education Status: B.S. in Software Engineering
- Skills: Python, Java, C++, SQL
- Frameworks: LangChain, LangGraph, PyTorch, FastAPI
- Interests: AI, Machine Learning, MLOps ,DevOps ,Cloud Computing
- Activities: Running

Summary

AI Engineer | Generative AI & Agentic Workflows
Specialized in architecting and shipping **production-grade LLM applications** using **LangGraph, LangChain and FastAPI**. Experienced in multi-agent systems, RAG and semantic recommendation (pgvector), text-to-SQL agents on BigQuery, and self-hosted LLM serving (vLLM). Proven track record of managing the full **MLOps lifecycle on GCP** (Cloud Run, Jenkins CI/CD), with a deep focus on deployment, LLM **observability** (Langfuse, LangSmith), and performance optimization.

Experience

08/25 - Present **AI Engineer** Zetaton

AI Patient Chat Coach & Recommendation Engine (digital health, US client) 03/26 - Present

- Shipped a **multi-agent LLM chat coach** (LangGraph, FastAPI) that is **live with real patients**, released through dev → staging → prod on **GCP Cloud Run**.
- Built long-term patient memory, **clinical safety guardrails with an LLM judge** on every reply, PII scrubbing (Presidio), and **multi-provider LLM fallbacks** (Groq, OpenRouter, Gemini).
- Designed a **semantic content recommendation engine** on Postgres/**pgvector** with context-aware queries and HyDE enrichment, raising retrieval accuracy.
- Instrumented the pipeline with **self-hosted Langfuse** (per-user cost, retrieval tracing), fixed a production segfault under scale-out, and built **Locust** load tests and LLM regression harnesses.
- Worked directly with **US stakeholders and clinical reviewers**, turning clinical feedback into prioritized engineering work.

AI Data Analysis Agent (retail shelf-intelligence client) 09/26 - 10/26

- Built an **LLM text-to-SQL agent on Google BigQuery** that answers business questions with charts, caveats and the exact queries used (production integration in progress).
- Built an **agentic data explorer** that profiles new client data, detects 12+ data traps and verifies AI-drafted meanings with SQL probes. Designed its port into a multi-tenant **LangGraph** platform, using Google's **BigQuery MCP server**.

Self-Hosted AI Coding Assistant (Trial) 09/26 - 10/26

- Deployed a **private, self-hosted LLM coding assistant** (Qwen3.8-27B on **vLLM**, NVIDIA RTX PRO 6000) so developers could use AI without sending source code to third parties.
- Built an **OpenAI-compatible API gateway** (LiteLLM) with per-developer keys and HTTPS (Caddy, Let's Encrypt), integrated with Aider, Cline and OpenCode.

Multimodal Recommendation System

- Developed a **Multimodal Recommendation System** leveraging Vision-Language Models (VLM) to provide context-aware suggestions based on user-uploaded images.
- Implemented **LangSmith** for real-time tracing.

07/25 – 09/25 **Data Science Intern** Paymob

- Engineered an **OCR-integrated chatbot** for financial document processing, utilizing **LangChain** and **LangGraph** to handle stateful extraction from Egyptian IDs and commercial records.

Education

2024 - 2025 **9-Months training Diploma - AI and ML Track** Information Technology Institute

2020 - 2024 **Bachelor's Degree, Computer Science** Cairo University

- Software Engineering Department (Grade : Very Good)

Projects

ANTIBIOTIC RECOMMENDATION & RAG EXPLANATION SYSTEM 06/2025 – 07/2025
Random Forest, XGBoost, local LLaMA model (mistral-7b-instruct-v0.1), ChromaDB, MLflow

- Combined ensemble ML and Retrieval-Augmented Generation (RAG) to predict suitable antibiotics for patients.
- Used MLflow to track and visualize experiments across models and parameters.
- Retrieved similar clinical cases with ChromaDB and generated explanations using a local LLaMA model.

BANK CUSTOMER TRANSACTIONS 04/2025 – 05/2025
Unsupervised Learning (K-Means, GMM), PCA, Silhouette Score, Elbow Method

- Identified meaningful customer segments through clustering.
- Used location embeddings + cosine similarity to reduce unique locations for better clustering.
- Gained insights into popular product combinations per group.

HYBRID-RECOMMENDATION SYSTEM 03/2025 – 04/2025
Collaborative filtering (SVD), Content-based filtering, Docker, Claw Cloud

- Built a client-server movie recommender (PyQt5 frontend + hybrid backend).
- Combined SVD collaborative filtering with cosine similarity content filtering.
- Containerized backend with Docker, pushed to DockerHub, and deployed on Claw Cloud.

CHURN PREDICTION MODEL DEPLOYMENT PIPELINE 05/2025
AWS (EC2, ECR), MLflow, Docker, GitHub Actions, Prometheus

- Built a churn prediction model using Logistic Regression.
- Containerized the model with Docker and deployed on AWS EC2 via ECR.
- Integrated Prometheus metrics and automated CI/CD with GitHub Actions.

SELA-app 02/2024 – 07/2024
C#, PostgreSQL, Swagger, Redis, Azure

- Developed a mobile app connecting individuals with organizations for online services.
- Deployed application on Microsoft Azure with Redis caching.

ONLINE-LEARNING-PLATFORM 01/2024 – 02/2024
Java, Spring Boot, Java EE, Microservices, Docker

- Engineered a learning platform with RESTful APIs.
- Designed microservices-based architecture, splitting the application into distributed parts.

Awards

2/25 **AWS Egypt Deep Racer League** NTI